CSC 34300
Lab Assignment 1

Due Date: September 27th 2024 before midnight

<u>Problem Statement:</u>

You are tasked with designing a control system for a 4-floor elevator using **combinational** logic. The elevator already has a controller circuit that manages the motor and direction of the elevator based on the input provided to it. However, the system is missing the input circuit that connects the floor buttons and provides the necessary signals to to the controller. Based on the information given to you, you know that:

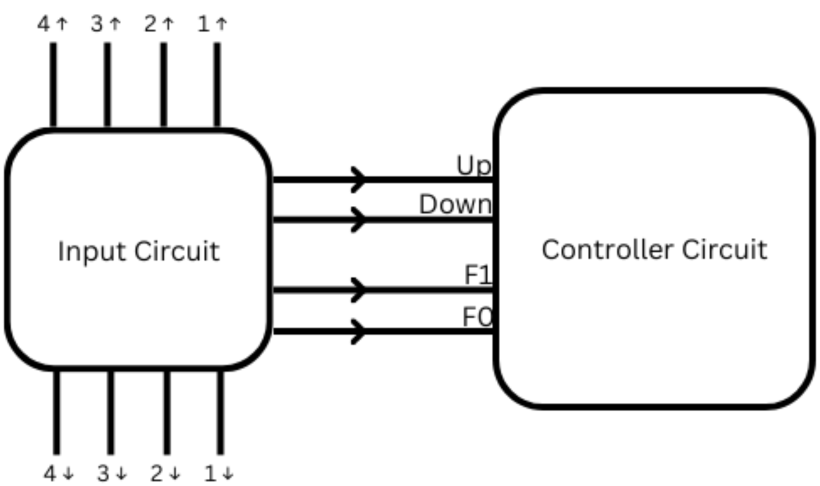
1. The controller circuit that already exists has 4 inputs.
2. You are required to design a logic circuit that supplies the inputs to the controller circuit. The circuit you're designing will have 8 inputs and 4 outputs.

To make the system more efficient, the following conditions must be met:

1. Your circuit can't request upward and downward movement together.
2. Priority is given to higher floors. If two or more buttons are pressed at the same time, your circuit should only consider the uppermost floor.
3. Priority is given to upward movement. If someone presses the up and down buttons together on the same floor, the system should decide to move upward.
4. For simplicity, assume that the maximum number of buttons clicked together at any time will be 2 (either up and down on the same floor, or two ups/downs on different floors).

Controller Circuit Specs:

The controller circuit already exists, you only interface with it and not implement it. The controller circuit has 4 inputs. In the block diagram below, they are given the names up, down, F1 and F0. The inputs up and down determine the direction of the elevator. They should **NOT** be HIGH at the same time. The inputs F1 and F0 represent a binary number of 2 bits to encode the floor number. Floors in the building are ground floor to 3rd floor corresponding to binary number 00 until binary number 11 respectively.



The inputs to the controller circuit are the outputs of the input circuit. You are required to implement the input circuit. The input circuit takes 8 inputs; 2 buttons for each floor of the building, and it controls 4 outputs; the inputs to the controller circuit. If none of the 8 buttons are pressed, this should result in a LOW up output and a LOW down output. The outputs for F1 and F0 aren't significant in this case. This is the idle state. If any of the 4 buttons for up is pressed, the output up should be HIGH and F1,F0 should have the floor number in binary. Similarly, if any of the 4 buttons for down is pressed, the output down should be HIGH and F1,F0 should also have the floor number in binary.

Deliverables:

Design the described logic circuit using **dataflow modeling only**. Simulate all possible test cases on ModelSim. Write a lab report and include the following:

- Your train of thoughts and how you reached the final implementation
- Two truth tables for the system (one with 4 ↑ buttons and the other with 4 ↓ buttons)
- Any inner signals you needed for your design
- Any boolean equations you derived
- Screenshots for some test cases on ModelSim
- Any challenges you faced

Save your report as PDF. Send **one** zip file that contains:

- Your PDF report.
- The VHDL file.

Name your zip file in the following format: {Last-name}_{First-name}_Lab1.zip.

Submit by email or on slack as DM to your lab instructor before the deadline. Undocumented late work will not be accepted.

Academic dishonesty is prohibited. Do **not** use help from any generative AI tools to complete this assignment.